

BUSINESS ANALYSIS REPORT

AdventureWorks

Global Sales, Customer & Returns Analysis — 2020 to Mid-2022

Prepared for: AdventureWorks Executive & Sales Leadership Team

Scope: 7 countries · 3 continents · 293 SKUs across Bikes, Accessories, and Clothing

Period Covered: January 1, 2020 – June 30, 2022

Report Date: July 2026

Executive Summary

This report analyzes AdventureWorks' global bicycle and accessories business across three sales years (2020–2022, through June) using 56,046 order lines spanning 25,164 unique orders, 17,416 customers, and seven countries. Total revenue over the period reached \$24.91M at \$10.46M in gross profit (a 42.0% blended margin), with 1,828 units returned across 84,174 units sold — a 2.17% return rate.

Bikes is overwhelmingly the core of the business, generating \$23.64M (94.9%) of total revenue and \$9.73M in profit, led by the Road Bikes and Mountain Bikes subcategories. Accessories, while contributing only \$907K in revenue, carries the highest margin in the portfolio at 62.8% — more than 20 points above Bikes' 41.1% margin — and drives disproportionate order volume (16,983 orders, more than Bikes despite the much lower revenue), making it a strong candidate for cross-sell and basket-building strategy.

Revenue grew consistently across the period, climbing from \$585K in January 2020 to \$1.83M in June 2022 — more than triple over roughly 2.5 years — with an especially sharp inflection beginning mid-2021. Geographically, North America (\$9.71M), Europe (\$7.79M), and the Pacific (\$7.42M, entirely Australia) are near-evenly split at the continent level, but at the country level Australia alone (\$7.42M) outperforms every individual European or North American market, and three U.S. territories (Southeast, Northeast, Central) show only a handful of orders each — a data coverage gap worth investigating before this is used as a growth benchmark.

Returns are concentrated in Bikes (3.08% return rate, roughly 50% higher than Clothing or Accessories), with several premium road and touring bike SKUs showing individual return rates above 5%. Customer revenue is broadly distributed rather than concentrated — the top 10 customers by revenue contribute roughly \$115K, well under 1% of total revenue — and skews toward Average-income (\$40K–\$80K) and Professional-occupation segments, though High-income customers convert to revenue at a notably higher rate per customer.

Methodology & Data Sources

This analysis consolidates three years of AdventureWorks order-line data (2020, 2021, and 2022 sales files) with five supporting lookup tables: Product (with Subcategory and Category joined in), Customer, Territory, Calendar, and a separate Returns table. Revenue and Profit were derived per order line as $\text{OrderQuantity} \times \text{ProductPrice}$ and $\text{OrderQuantity} \times (\text{ProductPrice} - \text{ProductCost})$ respectively, then aggregated across the merged model.

One data quality issue was identified and corrected during preparation: the Customer_Lookup file contained six non-data footer rows (an export timestamp and source label appended after the last real customer record) which were removed before analysis to avoid corrupting customer counts and segment totals — a good example of the kind of cleaning step worth documenting in a Power BI Power Query M script for reproducibility.

Findings reconcile closely against an in-progress Power BI dashboard build for this project: the dashboard's Customer Detail page independently shows 17.4K unique customers and \$1,431 revenue per customer, and its Exec Dashboard page shows \$24.9M revenue, \$10.5M profit, 25.2K orders, and a 2.2% return rate — all matching this report's calculations, and the Orders by Category figures (Accessories 17.0K, Bikes 13.9K, Clothing 7.0K) match exactly as well.

Table	Records	Key Fields	Role
Sales (2020–2022)	56,046	OrderDate, ProductKey, CustomerKey, Quantity	Fact table
Product Lookup	293	Category, Subcategory, Cost, Price	Dimension
Customer Lookup	18,148*	Income, Occupation, Education, Gender	Dimension
Territory Lookup	10	Region, Country, Continent	Dimension
Returns Data	1,809	ReturnDate, ProductKey, Quantity	Fact table

*After removing 6 non-data footer rows from the source file.

Sales Performance Overview

Headline KPIs across the full analysis window, January 2020 – June 2022.

REVENUE	GROSS PROFIT	ORDERS	CUSTOMERS	RETURN RATE
\$24.91M	\$10.46M	25,164	17,416	2.17%
2020–2022 YTD	42.0% margin	56,046 order lines	\$1,431 / customer	1,828 of 84,174 units

Category & Subcategory Performance

Bikes dominate the revenue base at \$23.64M (94.9% of total), split across Road Bikes (\$11.29M), Mountain Bikes (\$8.58M), and Touring Bikes (\$3.77M). Accessories generate just \$907K in revenue but at a 62.8% margin — more than 20 points higher than Bikes' 41.1% — while Clothing sits in between at 44.3% margin on \$365K in revenue.

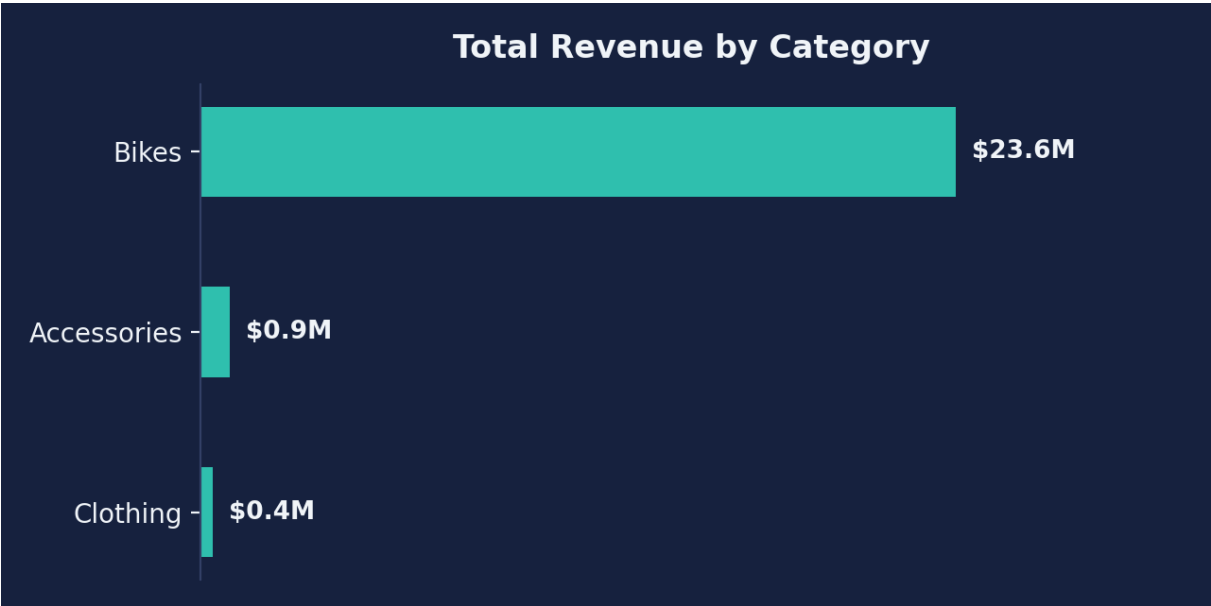


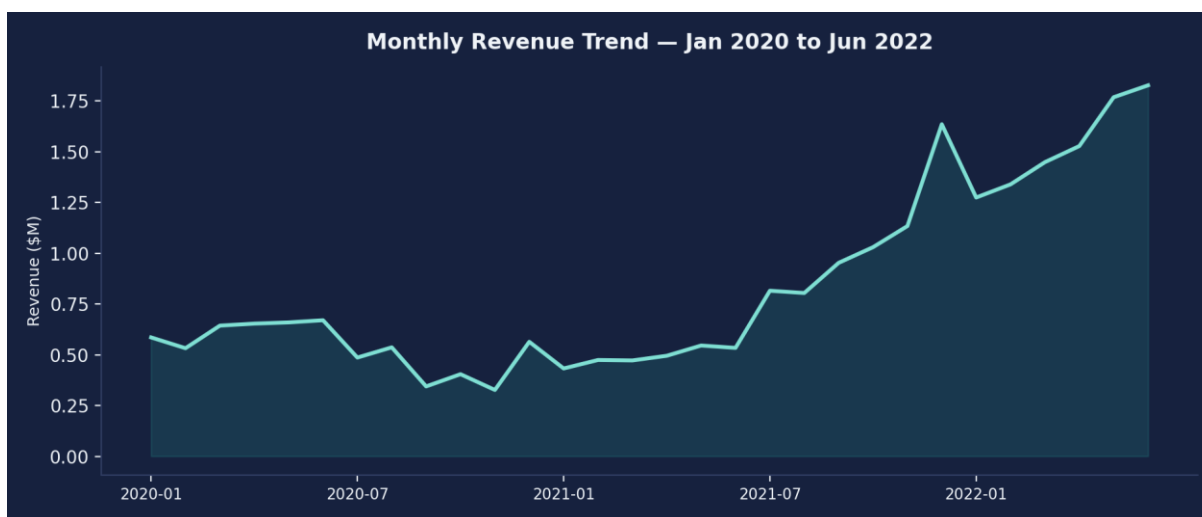
Figure 1. Total revenue by category, full period.

Notably, Accessories generates more individual orders (16,983) than Bikes (13,929) despite a fraction of the revenue — customers are clearly buying accessories both as bike add-ons and as standalone low-ticket purchases. Within Bikes, Mountain Bikes carries the strongest margin (45.8%) of the three subcategories, ahead of Road Bikes (38.7%) and Touring Bikes (37.8%), making it worth highlighting in merchandising even though Road Bikes has the larger revenue base.

- Accessories is the strongest margin-per-dollar category in the portfolio and a natural focus for bundle/cross-sell promotions at bike checkout.
- Mountain Bikes is the most profitable bike subcategory on a margin basis, not Road Bikes — worth reflecting in sales incentive structuring.
- The six best-selling individual SKUs are all Mountain-200 variants (Black/Silver, sizes 38–46), together generating over \$7.1M — a concentration worth monitoring for supply chain risk.

Revenue Trend & Growth Trajectory

Revenue grew steadily and consistently across the full period, from \$585K in January 2020 to \$1.83M in June 2022 — more than tripling over roughly two and a half years. Growth was gradual through 2020–2021 before a clear inflection point in mid-2021, after which monthly revenue climbed sharply and has sustained that higher growth rate through the most recent data (June 2022).

*Figure 2. Monthly revenue trend, January 2020 – June 2022.*

This sustained, accelerating trend line — rather than a one-time spike — suggests durable demand growth rather than a seasonal or promotional artifact, and provides a strong baseline for forecasting in the next phase of the Power BI build (e.g., a trend line or forecast visual on the Exec Dashboard page, consistent with the dotted trend overlay already present in the dashboard screenshots).

- The mid-2021 inflection point is worth investigating against any known product launches, market expansions, or pricing changes from that period to understand what drove the acceleration.
- Given the consistency of the growth trend, a rolling 12-month forecast measure would be a natural addition to the Power BI model.

Geographic Performance

At the continent level, revenue is nearly evenly split between North America (\$9.71M), Europe (\$7.79M), and the Pacific (\$7.42M, entirely Australia). But at the country level, Australia alone (\$7.42M) outperforms every individual North American or European market — including the United States as a whole, once broken into its constituent territories.

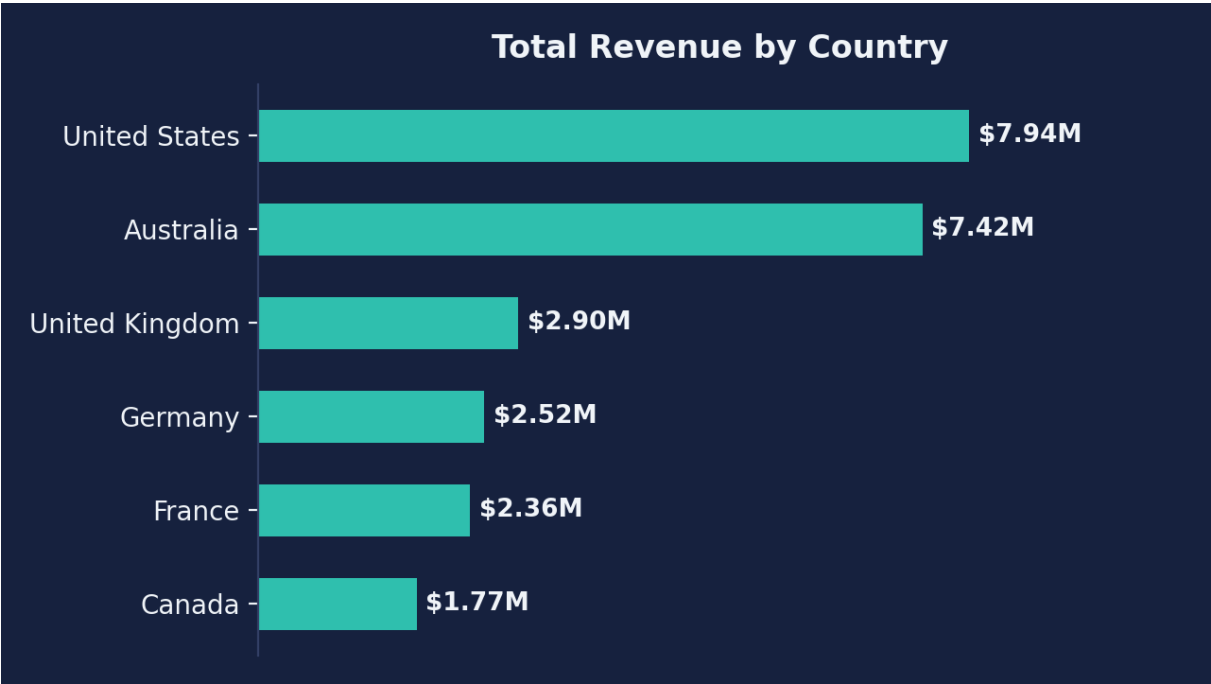


Figure 3. Total revenue by country, full period.

A data coverage gap is worth flagging here: three U.S. territories — Southeast, Northeast, and Central — show only 9 to 14 total orders each across the entire period, compared to thousands of orders in Southwest and Northwest. This is almost certainly a sales-coverage or data-completeness issue (e.g., those territories may not have been actively selling, or records may be incomplete) rather than genuine underperformance and should be confirmed with the business before being presented as a true regional comparison on the dashboard.

- Australia's outsized performance relative to any single European or North American country is worth a dedicated dashboard callout — it is currently AdventureWorks' single strongest market.
- Before publishing a country/region comparison visual, add a filter note or data-quality caveat for the three thin U.S. territories to avoid misleading conclusions.

Returns Analysis

The overall return rate of 2.17% is healthy for a bike and accessories retailer, but it is not evenly distributed. Bikes carry the highest category return rate at 3.08% — roughly 50% higher than Clothing (2.16%) and Accessories (1.95%) — which is a meaningful signal given Bikes is also 95% of total revenue.

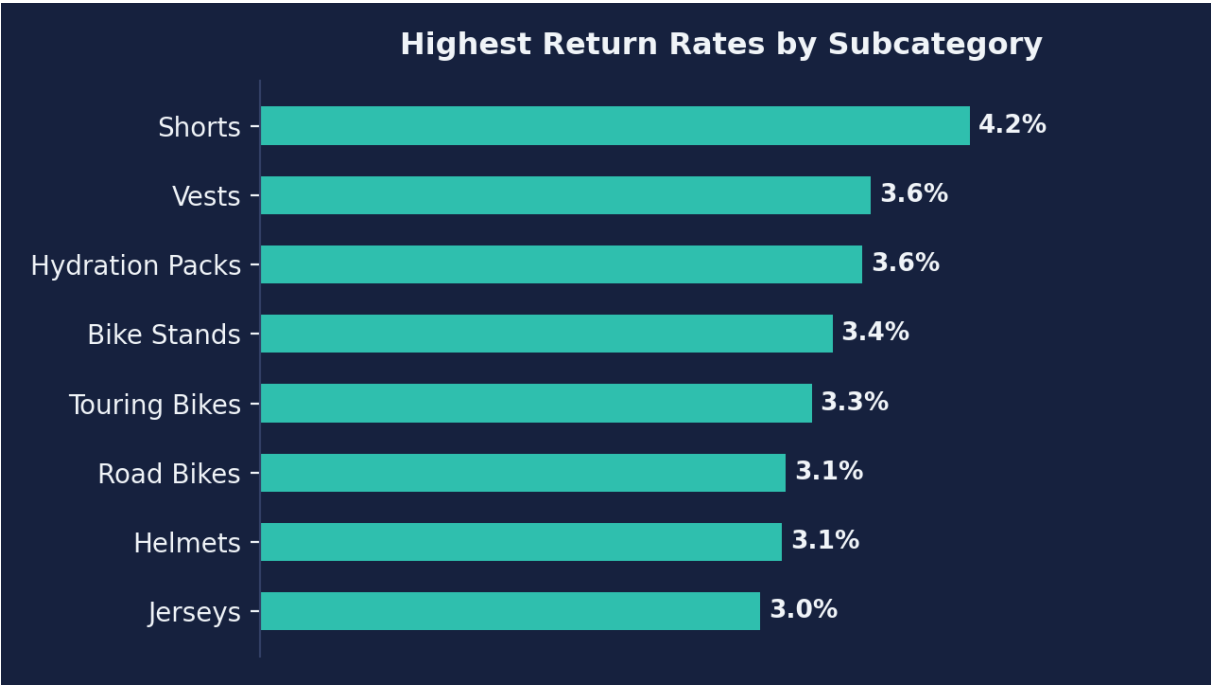


Figure 4. Highest return-rate subcategories (minimum 50 units sold).

At the individual product level, several premium road and touring bike SKUs show return rates well above the category average — the Road-650 Red (52cm) tops the list at 11.8%, followed by several Touring-2000 and Touring-3000 variants above 5%. These are lower-volume, higher-price SKUs, so a handful of returns moves the rate significantly, but the concentration in road/touring bikes (rather than being spread evenly across all bike types) is worth a closer look at sizing charts, product descriptions, or QC for that specific line.

- A product-level return-rate table filtered to Bikes, with a minimum-volume threshold to avoid small-sample noise, would be a valuable addition to the Product Detail page already in the dashboard.
- The Road-650 and Touring-2000/3000 lines are the strongest candidates for a root-cause review — sizing, description accuracy, or QC — given their above-average return concentration.

Customer Segmentation

Customer revenue is broadly distributed rather than concentrated in a small group — the top 10 customers by revenue contribute roughly \$115K combined, well under 1% of total revenue, indicating a healthy, non-dependent customer base rather than a handful of accounts driving the business.

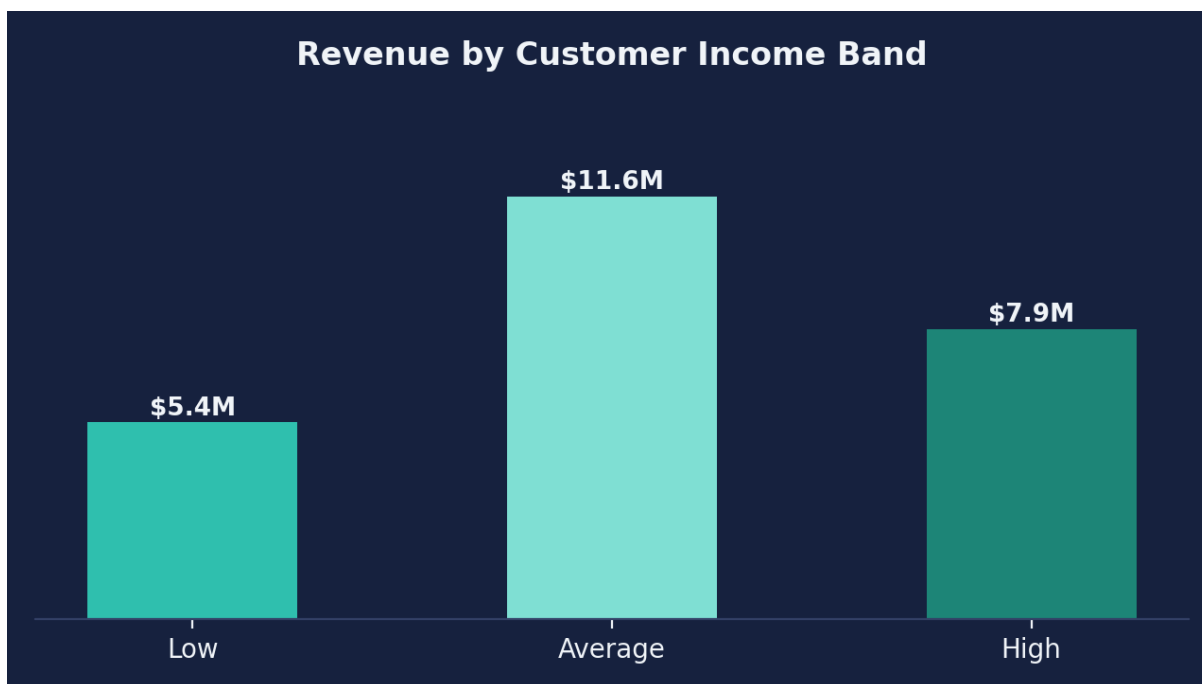


Figure 5. Revenue by customer income band.

By income band, Average-income customers (\$40K–\$80K) generate the most total revenue (\$11.58M from 8,339 customers), simply reflecting that this is the largest segment. High-income customers (4,155 people) generate \$7.95M — a notably higher per-customer average than either other band — suggesting they are the more valuable segment on a unit-economics basis even though they're not the largest group. By occupation, Professional customers lead on both revenue (\$8.47M) and customer count (5,219), followed by Skilled Manual and Management.

- High-income customers convert to revenue at a meaningfully higher per-capita rate — a strong segment to target in any loyalty or premium-product marketing push.
- The broad, non-concentrated customer base is a structural strength: no single customer or small group represents a meaningful revenue-concentration risk.

Key Findings

- Bikes drive 95% of revenue, but Accessories is the true margin engine at 62.8% — more than 20 points above Bikes — and generates more orders despite a fraction of the revenue.
- Revenue has more than tripled since January 2020, with a clear, sustained acceleration beginning mid-2021 — a genuine growth trend, not a seasonal spike.
- Australia is AdventureWorks' single strongest country market, outperforming every individual European or North American country.

- Three U.S. territories (Southeast, Northeast, Central) show a near-total absence of order data — a data coverage issue to resolve before using territory-level U.S. comparisons in the dashboard.
- Bikes carry the highest return rate (3.08%) of any category, with several premium road/touring SKUs individually exceeding 5–11%, worth a targeted quality or sizing review.
- The customer base is healthily distributed with no revenue concentration risk, and High-income customers convert at a notably higher per-customer rate — the strongest segment for premium targeting.

Strategic Recommendations

1. Build an Accessories cross-sell push at bike checkout

Given Accessories' 62.8% margin and high order frequency, a 'complete the purchase' bundle prompt at bike checkout — helmets, bottles, tire kits — is a low-cost way to lift blended margin without discounting the core bike business.

2. Investigate and resolve the U.S. territory data gap

Before the Power BI dashboard's Map or Territory visuals go into a stakeholder-facing state, confirm with the sales operations team whether Southeast, Northeast, and Central U.S. territories are genuinely inactive or whether this reflects incomplete data capture — the current numbers are not reliable for a regional performance conversation.

3. Run a quality/fit review on the Road-650 and Touring-2000/3000 lines

These specific SKUs show return rates 2–5x the Bikes category average. A focused review of sizing charts, product photography, and QC on this line could meaningfully reduce category-level returns given how concentrated the issue is.

4. Build a High-income customer segment view

Given their stronger per-customer revenue conversion, a dedicated Power BI segment (income band × product category affinity) would help target premium product launches and loyalty offers more precisely than a blanket campaign.

5. Add a rolling revenue forecast to the Exec Dashboard

The consistency of the growth trend from 2020–2022 supports a statistically grounded forecast measure (e.g., linear or exponential trend) as a natural next addition to the existing Revenue Trending visual, giving leadership a forward-looking view alongside the historical trend line already in place.

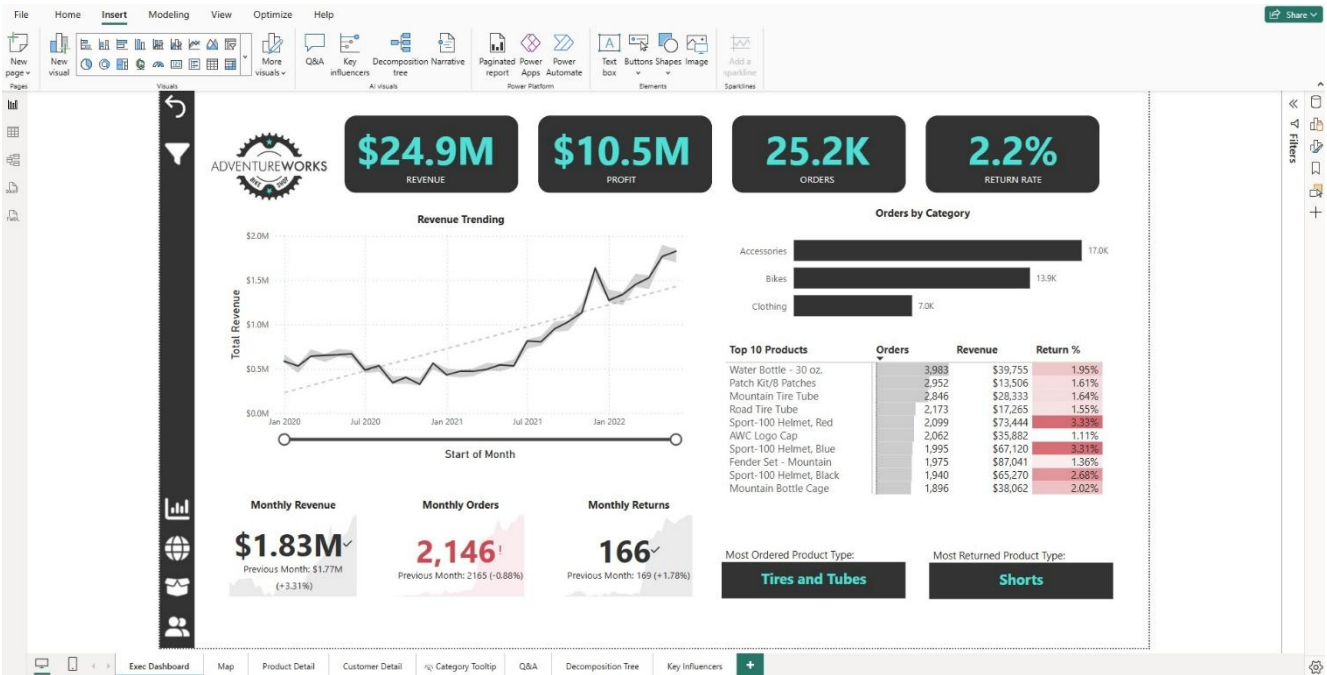
Appendix: Category & Subcategory Summary

Full-period revenue, profit, and margin by category and top subcategories.

Category / Subcategory	Revenue	Profit	Margin
Bikes (total)	\$23,642,500	\$9,726,168	41.1%
— Road Bikes	\$11,287,180	\$4,368,347	38.7%
— Mountain Bikes	\$8,583,748	\$3,930,662	45.8%
— Touring Bikes	\$3,771,565	\$1,427,160	37.8%
Accessories (total)	\$906,673	\$569,760	62.8%
Clothing (total)	\$365,419	\$161,787	44.3%

Data Sources: AdventureWorks_Sales_Data_2020/2021/2022.csv (56,046 order lines), AdventureWorks_Product_Lookup.csv (293 SKUs), AdventureWorks_Customer_Lookup.csv (18,148 records), AdventureWorks_Territory_Lookup.csv (10 territories), AdventureWorks>Returns_Data.csv (1,809 return records).
Period: January 1, 2020 – June 30, 2022.

Appendix: Executive Dashboard



Appendix: Customer Detail Dashboard

